

Enhanced Visual Question Answering for Preschool Education with ViLT, BLIP-2, and Sinkhorn Distance

Chengzhu Huang

School of Computer Science, The University of Sydney

In the field of machine learning, image recognition remains a significant challenge. This project focuses on a Visual Question Answering (VQA) system designed to interpret and answer image-based questions suitable for preschool children. The motivation behind this research is to foster children's cognitive development through advanced AI technology, enabling the system to comprehend and respond to inquiries regarding fundamental concepts such as counting, color recognition, and geometric shapes.

Keywords: Transformer | ViLT | BLIP-2

This project utilizes Transformer-based models, specifically ViLT and BLIP-2, to address this problem. ViLT integrates visual and textual inputs simultaneously, employing an attention mechanism to align features across the two modalities. BLIP-2 enhances the system by generating descriptive and contextually relevant captions. Key techniques include the implementation of a consistency classifier to ensure semantic relevance and the application of Sinkhorn Distance to optimize feature alignment, which collectively improve the system's accuracy.

The study achieved significant improvements in both image caption accuracy and the ability to accurately answer visual questions. The ViLT model demonstrates high average accuracy and F1 scores, indicating its effectiveness in integrating visual and textual information. The introduction of BLIP-2 further augments the text description capabilities. The results highlight the potential of Transformer-based models in preschool education. Although the project encountered challenges such as dataset limitations and computational requirements, it underscores the necessity for further optimization and exploration of model fusion techniques. Future research should focus on expanding dataset diversity, refining model architecture, and exploring server-based resources to overcome hardware limitations.

Section One: Introduction

Visual Question Answering (VQA) is a complex task requiring the integration of Computer Vision (CV) and Natural Language Processing (NLP). This project explores the Transformer model to resolve image comprehension issues for preschoolers, focusing on basic cognitive tasks such as counting, color recognition, and geometric figure identification. The significance of this research lies in its potential to enhance educational technology, particularly in preschool education, by providing tools that facilitate learning through interactive visual questioning.

1.1 Review of Previous Work. Significant advancements have been made in VQA research, primarily through the integration of text and image interpretation utilizing deep learning models. However, due to the distinct cognitive requirements of young children and the inadequacies of current models in addressing basic intuitive reasoning, implementing these strategies in preschool education remains difficult. The goal of this project is to bridge these gaps by developing a Transformer-based methodology specifically tailored for VQA activities at the preschool level.

1.2 Research Objectives. The primary objective of this research is to develop a VQA system for preschool image recognition capable of understanding and answering relevant questions. Specific goals include enhancing the ViLT model to handle preschool visual tasks, compiling and modifying datasets, improving model components to optimize performance, adjusting evaluation criteria to better reflect preschool children's understanding, and optimizing computational efficiency for real-time responses.

1.3 Methodology Overview. This project employs a combination of ViLT and BLIP-2 models to handle VQA tasks. The ViLT model utilizes a transducer-based architecture to process visual and textual inputs, enhancing feature extraction and the understanding of image semantics. The BLIP-2 model's Q-Former technology further improves the accuracy and detail of image descriptions. Key technologies include image-text matching to ensure precise alignment of text descriptions with visual data, consistency classifiers to maintain semantic consistency between generated text and visual content, and the Sinkhorn Distance method to optimize the alignment of text and image features, thereby enhancing multimodal learning.

1.4 Outline of the Paper. The paper begins with an introduction outlining the problem domain, its significance, and the research objectives. The Related Literature section reviews key studies, methods, and findings in VQA, highlighting the gaps this study aims to address. The Project Question section details the purpose, objectives, and guiding questions of the study. The methodology, data gathering, analysis, deployment, and testing procedures are covered in the Methods section. The project's materials, software, hardware, and roles are described in the Resources section. Experimental results are presented and discussed in the Results section. These results are interpreted, their implications for the field discussed, and contrasted with earlier research in the Discussion section. Finally, the section titled "Limitations and Future Work" delves

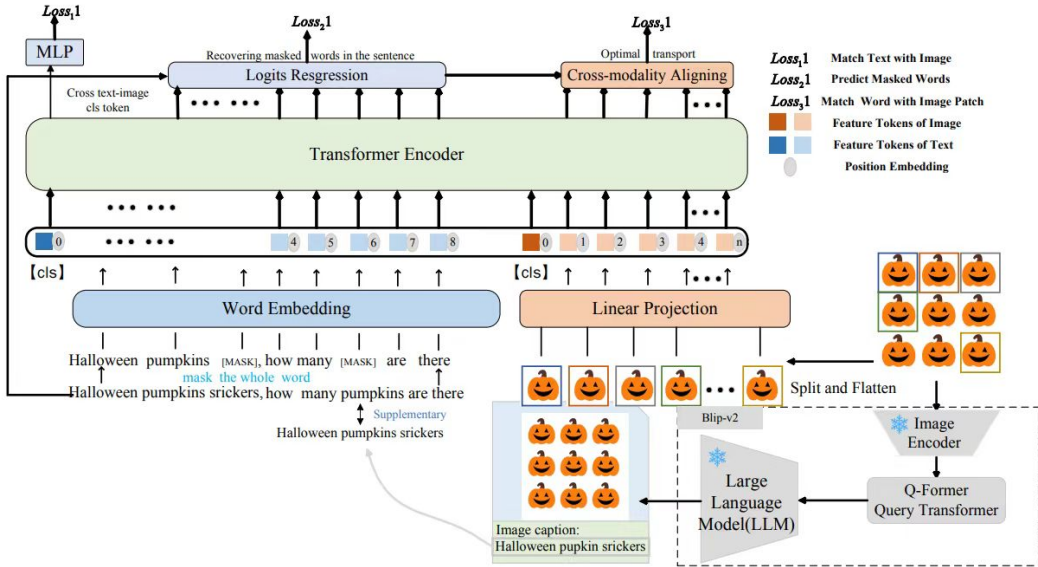


Fig. 1. Architecture

into the obstacles faced and potential avenues for further investigation. References to cited works and related literature are included at the end of the study.

Section Two: Related Work

Globally, approximately 60% of children aged three to six are enrolled in preschool education. However, in low-income countries, only one in five children attend preschool. Early education has a profound effect on children’s brain development. To improve the quality of preschool education, Visual Question Answering (VQA) can be utilized as an assistive tool [2].

Visual Question Answering is a field that combines Computer Vision and Natural Language Processing. In recent years, with the development of deep learning, the field of VQA has made remarkable progress. Image recognition is crucial in children’s education; however, applying VQA to image understanding for preschool children faces challenges. Children exhibit distinct behavioral characteristics and cognitive limitations, and their understanding of images relies more heavily on visual characteristics, necessitating analysis different from that used for adults [10]. To better understand the role of VQA in this domain, our project aims to design an image question answering system for children by using a Transformer model combined with Computer Vision and Natural Language Processing technologies. Therefore, we analyze early research related to VQA, focusing on the use of Transformers and other deep learning technologies.

VQA involves both image and text understanding, and many researchers have employed deep learning models to solve VQA problems. Regarding the strengths, weaknesses, and future directions of VQA, [12] conducted a comprehensive review of VQA models, discussing simple joint embedding models, attention models, knowledge integration models, and

domain-specific models. This analysis aids in understanding the characteristics of different VQA models and their potential application in preschool image question answering systems. [1] proposed the VQA dataset and baseline model, laying the foundation for this field. Additionally, [13] introduced a general-purpose visual-language representation pre-training method called VL-BERT, which uses BERT models for feature extraction. BERT processes text sequences through a bidirectional Transformer structure to generate context-sensitive embedded representations. Images and text are processed by ResNet and BERT, respectively, to extract visual and textual features, which are then input into a multi-layer Transformer structure for fusion. This improves visual commonsense reasoning and VQA task performance by combining visual and linguistic embedding features. Subsequently, researchers studied various methods to improve model efficiency, such as using hybrid LSTM-RNN with deep features for image description and using Transformers to generate image-language descriptions [3]. This research is particularly relevant for tasks such as dialogue systems, image annotation, and VQA, highlighting the importance of accurate image and language translation. [17] further enriched research in this area by proposing a method for discovering rich visual cues by mining question-answer prompts. This utilizes multimodal large language models (MLLMs) to generate and encode image-related question-answer pairs to improve the model’s reasoning ability. This approach is particularly useful for VQA tasks requiring complex chains of world knowledge and reasoning, such as OK-VQA and A-OKVQA. Experimental results show that Q&A Prompts achieve significant performance improvements on these challenging datasets compared to existing state-of-the-art methods.

The Transformer model was originally proposed by [16] for natural language processing tasks. The "attention mechanism" introduced by [16] replaces the traditional recurrent neural network, making the model more efficient in processing long input sequences. Due to its powerful parallel processing capability and ability to capture long-distance dependencies,

Transformer models have been rapidly applied in many fields. [9] investigated integrating ResNet and Transformer models to process image input more efficiently. This study highlights the Transformer’s critical role in enhancing visual recognition and reasoning capabilities in VQA tasks. By utilizing a unified neural architecture, improved performance in understanding and interpreting visual content was demonstrated, which is essential for developing VQA systems for preschool education. In the field of multimodal deep learning, the introduction of Transformers and attention mechanisms represents a major breakthrough. [14] provided an in-depth analysis of the current state of multimodal deep learning and its potential to advance VQA systems, exploring various approaches and their applications, and highlighting how these techniques can improve the integration of visual and textual information for more accurate, context-aware VQA models.

The cognitive ability of preschool children differs from adults; they are more inclined towards intuitive perception and pay more attention to specific image features. Preschool children show strong intuitive learning abilities in color recognition, shape recognition, and object counting [11]. Therefore, image question answering systems for preschool children must consider these specific needs. A VQA system based on a Transformer model can provide accurate answers and interactive experiences through efficient fusion of image and text features [8]. The ViLBERT model proposed by Kim et al. shows excellent performance on visual and linguistic tasks. Additionally, [18], based on the deep-stacked attention network model, aimed to enhance human-computer interaction by enabling the model to learn and reason about visual content. This model focuses on relevant parts of images when answering questions, thus improving accuracy and interpretability. This approach is particularly beneficial for educational tools because it ensures that the system can effectively understand and respond to children’s queries about images.

In the field of VQA, the BLIP-2 model plays a significant role. [6] introduced a new approach to visual and language pre-training. BLIP-2 utilizes frozen image encoders and large language models to significantly reduce training costs and complexity through a two-stage pre-training strategy: visual-language representation learning by the image encoder, followed by visual-language generation learning by the language model. BLIP-2 achieves state-of-the-art performance on multiple visual-language tasks, such as outperforming Flamingo 80B on the zero-shot VQAv2 task with 54 times fewer training parameters. This approach offers important implications for the development of efficient, low-cost image understanding systems for preschool children.

Overall, VQA has made significant progress in image recognition and understanding, but challenges remain in applying VQA technology to preschool education [4]. Our Transformer-based image question answering system aims to help preschool children better understand and learn from visual information.

Section Three: Proposed Methods/Framework

3.1 Methods. Our proposed methodology integrates three principal innovations to solve image understanding problems

for preschool children through a Visual Question Answering (VQA) system. These innovations include (1) the BLIP-2 model for advanced image-text interaction, (2) a Consistency Classifier to ensure semantic relevance, and (3) the use of Sinkhorn Distance for optimal feature alignment. This integrated approach aims to enhance the system’s accuracy in understanding and responding to visual queries posed by children. The framework’s processing flow begins with the input image being processed by the BLIP-2 model, which generates descriptive text corresponding to the image content. This text is then fed into the Consistency Classifier, which assesses the accuracy and relevance of the generated text to the image. Once the text is confirmed to align closely with the image, Sinkhorn Distance is used to calculate and optimize the alignment between text and image features, as shown in Figure 1. This innovative feature alignment method enhances the model’s performance in multimodal learning, improving the relevance and accuracy of responses.

BLIP-2 Model for Image-Text Interaction. The incorporation of the BLIP-2 model leverages its advanced language-image pre-training capabilities to generate precise and contextually rich image descriptions. These descriptions are pivotal in enhancing the semantic linkage between images and corresponding textual data. By creating captions that contextualize visual elements, the BLIP-2 model provides a comprehensive semantic layer instrumental for engaging in deeper interactions with query texts. This is especially beneficial for handling complex queries posed by preschool children, where understanding subtle visual cues can significantly influence response accuracy. By synthesizing descriptive captions, the BLIP-2 framework effectively primes the model for subsequent processing stages, ensuring that both initial perception and resultant textual output are deeply interconnected.

Consistency Classifier. The introduction of the Consistency Classifier is crucial for ensuring that textual output generated from image descriptions is not only grammatically coherent but also semantically aligned with visual data. This alignment is critical for maintaining high information fidelity across modalities, particularly in educational settings where content accuracy is paramount. The classifier employs a dense neural network layer followed by dropout and sigmoid activation to assess the probability that the text accurately reflects the image’s semantic content. This approach is underpinned by the need to maintain educational content integrity, ensuring generated question-and-answer pairs are both informative and reflective of the visual stimulus.

Sinkhorn Distances for Feature Alignment. The utilization of Sinkhorn Distance for feature alignment serves to quantify and minimize discrepancies between text and image features. This optimal transport metric allows for precise adjustments of multimodal features, thereby enhancing the model’s capability to handle complex visual and textual data. The iterative nature of the Sinkhorn algorithm plays a crucial role in refining these alignments, which is fundamental for the accuracy and efficiency of the multimodal learning process. To align text and image features, we employ the Sinkhorn Distance, which involves the following steps:

- (1) **Compute the Cost Matrix:** The cost matrix $\mathbf{C}$ is calculated as the squared Euclidean distance between the

feature vectors from text $\mathbf{x}$ and image $\mathbf{y}$:

$$C_{ij} = \|x_i - y_j\|^2$$

This matrix is normalized to stabilize the computations.

- (2) **Kernel Matrix:** The kernel matrix $\mathbf{K}$ is derived from the cost matrix using an exponential transformation with a regularization parameter ϵ :

$$K = \exp\left(-\frac{C}{\epsilon}\right)$$

- (3) **Sinkhorn Iterations:** The algorithm iteratively updates the log-scaling vectors to compute the transport plan $\mathbf{T}$, which represents the optimal alignment between text and image features:

$$T = \exp(\log \mathbf{u} + \log \mathbf{K} + \log \mathbf{v})$$

- (4) **Wasserstein Distance:** The Sinkhorn Distance, or approximate Wasserstein Distance, is then computed as:

$$\text{Sinkhorn Distance} = \sum_{i,j} T_{ij} C_{ij}$$

By incorporating Sinkhorn Distance, our model aligns text and image features more accurately. This alignment is critical for accurate and meaningful multimodal interactions, essential for tasks such as visual question answering. Iterative optimization ensures that textual and visual representations are well-aligned, enhancing the model’s ability to understand and respond to complex multimodal data. This method effectively reduces dissimilarity between text and image features, optimizing alignment for improved multimodal learning performance.

3.2 Model Architecture.

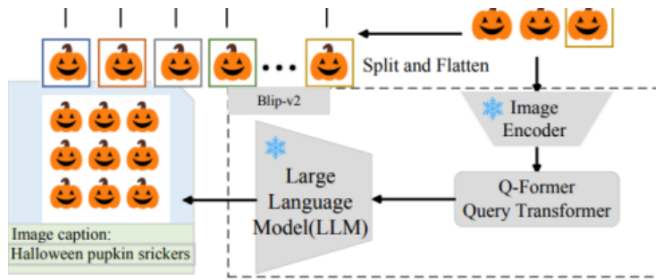


Fig. 2. BLIP-2 Model

BLIP-2 Model. As the figure shows, the model architecture of BLIP-2 comprises components forming the foundation language model.

This figure demonstrates the combination of BLIP-2 and a Large Language Model (LLM) for processing and understanding image data. First, multiple input images (e.g., Halloween pumpkin stickers) are segmented and flattened by BLIP-2 for data encoding. Immediately after, key visual information is extracted using an image encoder. This information is then fed into a large language model responsible for generating textual descriptions matching the image content. Finally, the Q-Former query converter further optimizes the data and outputs the final image description.

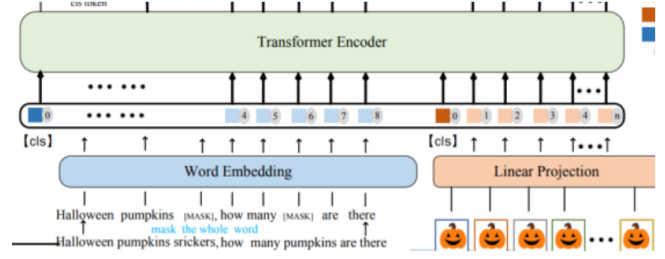


Fig. 3. Input Model

Input. As the figure illustrates, the input architecture is divided into components central to the language model.

Based on VQA requirements, the input is divided into a text part and an image part. The text part splits the sentence into serial word sequences. Converting text into embedded features (Word Embedding) usually involves splitting words, finding word embeddings, and adding positional encoding. The following details how to convert the sentence “How many pumpkins are there?” into an embedded feature:

1. **Segmentation:** The sentence is broken into words or subwords. For example, “How many pumpkins are there?” is split into [“how”, “many”, “pumpkins”, “are”, “there”].
2. **Finding word embeddings:** Each word is converted into an embedding vector using a pre-trained embedding matrix. In our approach, BertEmbedding is used.

In BertEmbedding, the initialized word embedding matrix embeds words (e.g., “pumpkins”) as vectors: “pumpkins” $\rightarrow$ [0.23, 0.15, -0.33, ..., 0.18], where the dimension is 768. ViLT adopts this approach, allowing our method to interact with both text and image information.

3. **Incorporating positional encoding:** To enable the model to understand word positions, positional encoding is added to each word embedding. Positional encoding is not added exclusively to image patches.

The image part is processed using 32×32 cut patches. Unlike the conventional ViT-B 16×16 size, we chose a larger patch size to mitigate the quadratic time complexity growth of the Transformer. A direct connection between text and image would increase complexity four-fold; therefore, larger patches reduce the number of image blocks (e.g., a 224×224 image yields 196 blocks with 16×16 cuts, but only 49 blocks with 32×32 cuts), thereby improving inference speed. These embedded image blocks are projected to a dimension of 768 sequences with added positional encoding. Thus, the input becomes a sequence of $1 + N \times 768$ and $1 + L \times 768$, where L is the number of words, N is the number of image blocks, and the added 1 represents the class token.

We utilize a class token, similar to the [CLS] token in BERT. ViT introduces this mechanism because the transformer input is a series of patch embeddings, and the output is a corresponding sequence of patch features. Ultimately, this must be summarized into a class judgment. While average pooling could summarize patch features, the original article appends a class token.

4. **Interaction:** Input unification facilitates self-attention operations through the transformer. Feature interaction relies purely on the QKV mechanism to establish links between text and image, realizing multimodal interaction.

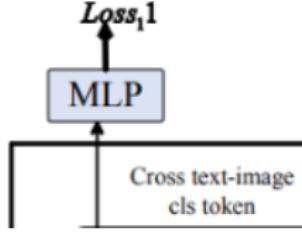


Fig. 4. Loss Function 1

Loss Function 1. To improve the interaction between language and textual information, we use Image Text Matching to enhance correspondence between textual images and construct feature interaction between images and textual tokens via matching. Specifically, we select a consistent image and text with a probability of 0.5 and determine whether the text and image match using Loss1 (a binary classification problem). An example follows:

For an image containing pumpkins, the question “How many pumpkins are there?” represents a matching sample. Simultaneously, we prepare a mismatching question: “How many pens are there?” We decide whether to use the corresponding question with a random probability of 0.5; if the corresponding question is used, the label is 1, otherwise, it is 0. The cross-entropy loss of the matching result is calculated based on the classifier’s output. Minimizing this loss optimizes text-image matching.

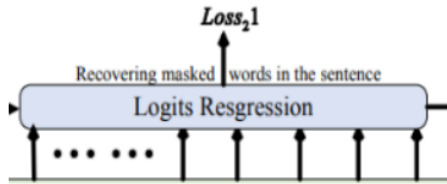


Fig. 5. Loss Function 2

Loss Function 2. The input is an $L \times 768$ text sequence. We mask words in the input sentence “How many pumpkins are there?” to create “How many *** are there?”. The task is to restore the semantic information of the masked word (e.g., pumpkins) using image information. The input comprises tokens corresponding to “How many *** are there?”, but after multimodal interactions, the output size remains $L \times 768$ (where $L = 5$ for the five words). The masked word’s vector can be inverted to “pumpkins”, and L1 loss is calculated against the original sentence “How many pumpkins are there?” to achieve optimal interaction effects.

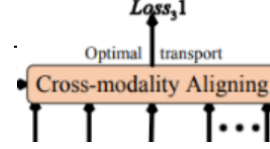


Fig. 6. Loss Function 3

Loss Function 3. The purpose of this step is to reduce the gap between image and text within the feature space, thereby improving the match between text and image. The inputs are the embedded features of the text and image. Since we are dealing with data disparities across different dimensional sizes, standard paradigms like L1-norm or L2-norm (which require uniform dimensions) are unsuitable. Therefore, Wasserstein distance minimization is calculated to characterize the difference between text and image.

Section Four: Experiments

4.1 Data. Our raw data is derived from the Pic2Question dataset. The authors divided the image data into two parts: SHAPE and OBJECT, corresponding to image category and shape, respectively. Additionally, the authors created quiz pairs for object image data with standardized answers. After downloading the dataset from GitHub and storing it via Google Drive, we initiated raw data processing.

Data Integration. In the initial phase of dataset preparation, we merged the two primary categories: SHAPE and OBJECT. This entailed meticulous pairing of image files with corresponding quiz questions. To ensure precise alignment, each pair was linked using unique identifiers (specifically image paths), as each image could encompass multiple question-answer contexts. Unlike conventional VQA datasets, ours was specifically tailored to meet the unique demands of this project, focusing on image-processing capabilities required for preschool educational content. Consequently, the dataset was structured to include descriptive text encompassing both image descriptions and associated question-answer pairs, alongside the image path. Importantly, all images were pre-standardized in size, obviating the need for additional resizing during preprocessing.

To augment the model’s access to contextual information, we employed advanced Python preprocessing techniques to refine the dataset. A significant aspect involved utilizing the BLIP-2 model to generate supplemental descriptions for each image. These descriptions were seamlessly integrated into our existing data structure, enriching the linguistic environment of the original quiz pairs. This enhancement was critical in aiding the model’s comprehension of visual content, bridging the semantic gap between text and image. Moreover, extensive preprocessing ensured optimal alignment of our dataset with the specific requirements of our VQA model, guaranteeing effective synchronization of multimodal inputs for efficient training. The strategic integration of these elements fosters deeper semantic interplay between textual and visual data, which is pivotal for the success of our VQA system in a preschool educational setting.

4.2 Data Working. During the processing of the Pic2Question dataset, we conducted rigorous cleaning to remove erroneous labels. This involved rectifying discrepancies between images and their corresponding quiz questions or generated descriptions, ensuring dataset integrity and consistency. For each image, we extracted relevant visual features using advanced image processing techniques. Textual features from questions and BLIP-2 generated captions were encoded using NLP methods. This dual approach ensured that both visual and textual data were optimally prepared for training.

Finally, the fully prepared dataset was divided into training, validation, and testing sets. This division was carefully balanced to ensure diverse and representative samples from both SHAPE and OBJECT categories across all subsets, facilitating unbiased training and evaluation.

4.3 Experiments settings. Our model has demonstrated remarkable robustness and accuracy in understanding and answering visual questions, with or without BLIP-2. However, we discovered an interesting result when including BLIP-2 generated captions in tasks involving simple image counting, particularly with children’s images. This inclusion led to a 1% decrease in accuracy, suggesting that while BLIP-2 is powerful, it may not enhance performance in every task.

Our dataset consists of preschool-level images, presenting unique challenges. To tackle this, our model is based on ViLT (Vision-and-Language Transformer), which is proven effective in understanding complex visual content. Our ablation study revealed key insights: the addition of BLIP-2 captions slightly decreased accuracy, indicating potential confusion rather than enhancement. We found that for simple image counting tasks, precise object detection and counting capabilities are more beneficial than detailed descriptions. This aligns with our observation that BLIP-2 captions introduced extraneous information, negatively impacting counting accuracy.

Hyperparameter analysis further illuminated how to optimize our model. Increasing epochs generally improves accuracy, with the highest accuracy achieved at 15 epochs. Additionally, a balanced learning rate of 0.000025 allows effective model convergence without overfitting. Two key hyperparameters, Theta and Beta, play crucial roles: Theta controls the weight of the Sinkhorn distance loss, ensuring effective alignment of text and image features; Beta controls the weight of the consistency classification loss, maintaining a balanced contribution to the overall loss.

In conclusion, our model shows impressive performance in understanding and answering visual questions, especially for preschool-level images. However, for simple image counting tasks, we must focus more on direct image feature extraction and object detection rather than relying on generated descriptions. By optimizing hyperparameters and understanding specific task requirements, we can continue to improve model performance and push the boundaries of VQA. We aim to develop a robust VQA system for preschool educational content. To achieve this, we used the ViLT model as a baseline and integrated the BLIP-2 model to improve performance. Detailed results of our experiments and evaluations follow.

Initially, we trained our VQA system using the ViLT model alone. The average accuracy was close to 99%, indicating the model provided correct answers for most test datasets. This reflects the ViLT multi-head attention mechanism’s ability to focus on multiple parts of the input data simultaneously to better understand image-text relationships. An average accuracy of 0.9888 signifies that most answers generated by ViLT are accurate. This demonstrates the model’s strong capability in feature extraction and multimodal information combination. The F1 score, a harmonic mean of accuracy and recall, is 0.9891, indicating a balanced overall performance. The model excels not only in generating accurate answers but also in covering all possible correct answers, demonstrating strong reliability for preschool VQA tasks.

Although the initial model performed well, we aimed to improve performance on specific tasks such as pattern recognition and counting, and to enhance stability. Therefore, we applied two innovative methods. Using the consistency classifier’s output and the target label to calculate loss improves robustness and performance under uncertainty. Results for ViLT-novelty show values slightly lower than the baseline ViLT model, though performance remains excellent. We attribute this to two potential reasons: First, consistency loss and optimal transmission loss may carry excessive weight during training, causing the model to prioritize these over the original task. Second, training on a single small dataset for too long may have induced overfitting.

In addition to internal adjustments, we investigated the BLIP-2 model, known for processing multimodal information, to address ViLT’s shortcomings in feature extraction and image processing. We added the BLIP-2 image encoder and Q-Former to the ViLT model to enhance image feature extraction. Training on the original test dataset (ViLT+BLIP-2) resulted in decreases across average accuracy, precision, recall, and F1 score. Although the decrease is minor, it indicates that the newly introduced modules and loss functions did not significantly improve performance under the current configuration and may have slightly hindered it. We believe this is because our preschool-focused data is relatively simple, containing less rich visual information. Consequently, Q-Former extracts fewer features, affecting performance in complex tasks. Models may also fail to fully leverage their complex structures, resulting in underutilized feature extraction capabilities.

We then fused the ViLT model with the added innovations conditional on the BLIP-2 model to leverage both improvements (loss adjustment + improved feature extraction), creating a model that better understands and processes multimodal data. Results for ViLT-novelty+BLIP-2 show slight improvements in overall accuracy (0.9897) and recall (0.9884) compared to ViLT+BLIP-2. After overall adjustments, the new model better captures key features when processing multimodal data, increasing correct recognition probability and helping the model better combine image and text information.

Comparing the four models, the baseline ViLT model exhibits the best performance in final data results. Its design and parameters are optimized for specific tasks like counting and shape recognition, reaching a balanced state. Adjusting the loss function and introducing new components (like BLIP-2’s

Model	Average Accuracy	Average Precision	Average Recall	Average F1-score
ViLT	0.9903	0.9888	0.9895	0.9891
ViLT_novelty	0.9889	0.9883	0.9879	0.9881
ViLT+BLIP-2	0.9893	0.9839	0.9878	0.9858
ViLT_novelty+BLIP-2	0.9897	0.9842	0.9884	0.9862

Table 1. Comparison of different models on various metrics

Q-Former) offers theoretical advantages but may introduce complexity and uncertainty, potentially leading to overfitting or underfitting given the existing data and tasks.

Method	with-VQG	with vqg-unknown	with Dual task	with Visual question	Das's result	vilt	vilt_novelty*
Accuracy	0.71	0.62	0.67	0.59	0.93	0.99158	0.98391

*vilt_novelty included(vilt, Blip-2, consistency classifier, sinkhorn distance)

Fig. 7. Comparison of different methods on accuracy

Comparison. According to [5], Visual Question Generation (VQG) is the task of generating relevant questions from input images, covering all aspects of the image. Based on VQG, [15] proposed VQG-unknown to obtain information about unknown objects. This method generates questions highlighting unknown or ambiguous aspects, enhancing anomaly detection. [7] improved overall performance by executing two related tasks simultaneously, exploiting shared knowledge when visual questions are generated or asked. [?] proposed a new approach using RCNN for object detection and localization, and LSTM to generate quiz pairs based on object descriptions. They extract features using VGG-16 during detection and process RCNN output to create textual image descriptions and generate quiz pairs.

Das's research direction is similar to ours, and our dataset is adapted from their Pic2Question dataset. Therefore, we compare our experimental results with theirs. Accuracy data shows that VQG-Unknown, Visual Question, and Dual Task have lower accuracy rates. Visual Question struggles with complex images due to a lack of advanced object detection and language modeling techniques. Dual Task performance is hindered by hardware limitations and the complexity of managing two tasks simultaneously. VQG-Unknown's focus on unknown elements may affect the generation of accurate questions about known objects, resulting in lower accuracy. The VQG goal is clearer; its ability to generate questions based on specific visual content, coupled with focused training, yields higher performance than VQG-Unknown.

Das combined advanced neural networks with a dataset tailored to preschoolers to achieve significantly improved accuracy. Their integration of RCNN, LSTM, and an object knowledge layer allows for precise question-answer generation, making their model effective for preschool assessment. Their methodology provides valuable references.

In this project, we use ViLT, which allows for more seamless integration of visual and verbal tasks than traditional CNN and RNN/LSTM-based models. The Transformer architecture effectively captures complex relationships between visual and

textual data. We further enhance the generation of consistent questions with BLIP-2. The consistency classifier ensures coherence and logical consistency with visual content. Due to advanced architecture and optimization techniques, we achieve higher accuracy than Das's experiments on the same dataset. While our model contains high-level components that improve performance, they also impact computational speed.

Table 2. Hyperparameter Research Results

Hyperparameter	Configuration	Average Accuracy
Default Parameters		
epoch	10	-
lr	0.0001	-
beta	0.001	-
theta	0.000005	-
Epoch and Learning Rate Case		
epoch	5, lr = 0.00001	0.97711
epoch	5, lr = 0.0001	0.99032
epoch	5, lr = 0.000025	0.98724
epoch	15, lr = 0.00001	0.97975
epoch	15, lr = 0.0001	0.98910
epoch	15, lr = 0.000025	0.98680
epoch	10, lr = 0.00001	0.98019
epoch	10, lr = 0.0001	0.98988
epoch	10, lr = 0.00005	0.99158
epoch	10, lr = 0.00007	0.98989
epoch	10, lr = 0.000025	0.98504
Theta Case		
theta	0.000001	0.95999
theta	0.00001	0.98151
theta	0.0000005	0.98988
Beta Case		
beta	0.0001	0.98768
beta	0.001	0.98988
beta	0.01	0.98011
beta	0.1	0.88468

Section Five: Results and discussion

5.1 Improvements in the Training Dataset. The dataset utilized for training is derived from the Pic2Question dataset, tailored to address preschool image understanding tasks. While this customization facilitates targeted learning, it limits the model's capability to generalize. The primary focus is on categorizing shapes and objects, with questions predominantly centered on counting colored objects.

Limitations include confinement to simple counting and color identification, which may not sufficiently challenge the

model. Images are primarily simple geometric shapes, not representing the variety of visual elements children encounter. The simplified tasks do not encourage the development of deep visual understanding. To address these limitations and enhance VQA effectiveness in preschool settings, we suggest the following improvements:

1. **Expansion of Question Types:** Introduce a wider variety of question types that go beyond simple counting and color identification. This could include questions about spatial relationships, comparisons (e.g., bigger/smaller, taller/shorter), and more complex attribute identification (e.g., identifying textures or patterns).
2. **Incorporation of Realistic Images:** To better simulate real-world conditions that children might encounter, the dataset should include more complex and realistic images. This includes pictures of everyday objects, scenes from children’s books, and illustrations that depict multiple objects interacting in various ways.
3. **Customization for Specific Cognitive Tasks:** As the project aims to assess specific capabilities such as counting ability, color concepts, knowledge of geometric shapes, and the perception of different objects by kids, it is imperative to create sub-datasets tailored to each of these tasks. This would allow for a more detailed analysis of the model’s performance in each domain and enable targeted improvements in its cognitive understanding capabilities.
 - (a) **Geometric Shapes:** Introduce more complex geometric figures and their combinations in real-life settings to deepen the model’s understanding of shapes.
 - (b) **Object Perception:** Create tasks that involve identifying objects in cluttered backgrounds or different states (e.g., a partially visible object), which mimics real-world visual perception challenges faced by children.

5.2 Discussion. Our project focuses on enhancing VQA models using Transformers. VQA is a complex task combining CV and NLP, requiring the model to process multimodal information, semantically align visual and textual features, and generate correct answers. Modeling was conducted for preschool children’s image recognition.

Initially, we found that the standard ViT model could not be directly applied to multimodal tasks efficiently. Therefore, we sought a Transformer model capable of processing both visual and verbal information while improving computational efficiency.

We transformed the ViT model into the ViLT model, which processes both image and text features within the same Transformer, enabling the understanding and generation of image descriptions. It reduces complexity by using image pixels directly as inputs, removing reliance on external feature extractors. To fine-tune the model, we introduced consistency classifiers and Sinkhorn Distance. These methods improve adaptation to various data formats, enhancing VQA accuracy and relevance.

A main challenge with ViLT was optimizing training efficiency and performance while handling multimodal tasks.

ViLT’s architecture handles complex scenes but consumes high computational resources. We simplified other model parts while maintaining key feature complexity. Specifically, we introduced cross-modal attention efficiency and different loss functions. This improves focus on visual-verbal tasks and simplifies the overall architecture, reducing unnecessary computational burden.

Additionally, we implemented stepwise learning strategies and data augmentation to utilize training data effectively and reduce overfitting. This enables ViLT to perform well in both single-modal and multimodal contexts. To improve visual information accuracy and innovativeness, we introduced BLIP-2. This process involves segmenting images for encoding, then inputting them into a Large Language Model (LLM) to generate textual descriptions. This enhances image content understanding and generates accurate textual information. Finally, Q-Former optimizes the data for specific queries. We tested four models using the same dataset. Although the fused model (ViLT + BLIP-2) performed slightly lower on some metrics than the separate ViLT model, it demonstrates higher flexibility and adaptability in handling multimodal information for complex tasks.

In summary, we made numerous adjustments to optimize performance. First, data consistency was ensured through preprocessing (e.g., normalization). Second, adjusting learning rates and regularization significantly improved training effects. Third, loss functions were adjusted for unbalanced data, and fusion strategies were tested. These adjustments enhance overall performance, stability, and robustness. We learned to balance model complexity and functionality, discovering that efficiency and results can coexist. Our enhancements to ViLT’s VQA response improved accuracy, highlighting the transformative potential of our work in preschool education. Looking ahead, we recommend promoting model fusion techniques, combining strengths of different models, adopting better data preprocessing, simplifying model structure to reduce training time, enlarging the dataset for extensive testing, and further optimizing the model for practical applications.

Section Six: Conclusion

6.1 Limitations. Despite the remarkable results achieved, several challenges regarding modeling remain.

First, dataset reliability presents concerns. While the Pic2Question dataset fit the project well, the lack of breadth and diversity in question pairs and image types may affect generalization to other preschool settings. Second, the massive sample size and limited hardware facilities restricted the scope and scalability of experiments, leading to extended training times. Advanced CPUs could save time, prevent time bias, and optimize the model. Third, the BLIP-2 model presents limitations. Its complex architecture (multiple attention mechanisms, multimodal fusion) requires high skill levels for proper implementation. The large number of hyperparameters makes tuning time-consuming. Furthermore, BLIP-2 requires significant computational resources, limiting its feasibility for teams with restricted resources and hampering optimization efforts. Fourth, personnel constraints (team size and technical

level) impacted the process. Increasing team size and technical expertise could improve efficiency. Fifth, time constraints may have affected further parameter tuning or architectural adaptation, potentially impacting overall effectiveness. Tight schedules and lack of preparation led to schedule deviations and insufficient optimization time.

6.1 Future work. Substantial scope for improvement remains. First, we can optimize the model by continuously adjusting parameters and increasing iterations. We must collect data more extensively and implement augmentation techniques to address diversity limitations. Testing on different datasets can greatly improve model breadth. Future work may include expanding the dataset via partnerships with educational institutions to improve robustness. Second, utilizing server-based resources can overcome hardware limitations, enabling the training of larger models. We can reduce bias through routine testing against different datasets. Ultimately, longitudinal studies tracking educational impact on preschool learners could provide valuable insights. Integrating existing platforms can expand the project's scope.

In conclusion, although our VQA system has been successful, development space remains. Our system is valuable not only in education but can be integrated into industries like medical image analysis (helping physicians via natural language questioning) or patient self-help systems. In the future, VQA technology can facilitate real-time Q&A in online education platforms, improving learning efficiency and effectiveness.

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References

- [1] Agrawal, A., Lu, J., Antol, S., Mitchell, M., Zitnick, C. L., Parikh, D., and Batra, D. (2016). Vqa: Visual question answering. *International Journal of Computer Vision*, 123(1):4–31.
- [2] Bank, W. (2017). Early childhood development.
- [3] Deorukhkar, K. P. and Ket, S. (2022). Image captioning using hybrid lstm-rnn with deep features. *Sensing and Imaging*, 23(1).
- [4] He, B., Xia, M., Yu, X., Jian, P., Meng, H., and Chen, Z. (2017). An educational robot system of visual question answering for preschoolers. In *2017 2nd International Conference on Robotics and Automation Engineering (ICRAE)*. IEEE.
- [5] Krishna, R., Bernstein, M., and Li, F. (2019). Computer vision and pattern recognition (cvpr). pages 2008–2018.
- [6] Li, J., Li, D., Savarese, S., and Hoi, S. (2023). Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models.
- [7] Li, Y., Duan, N., Zhou, B., Chu, X., Ouyang, W., Wang, X., and Zhou, M. (2018). Visual question generation as dual task of visual question answering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 6116–6124.
- [8] Lu, J., Batra, D., Parikh, D., and Lee, S. (2019). Vilbert: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks.
- [9] Luo, C., Gong, B., Chen, T., and Sun, C. (2023). Towards a unified neural architecture for visual recognition and reasoning.
- [10] Ni, Q. and Yu, Y. (2015). Research on educational mobile games and the effect it has on the cognitive development of preschool children. In *2015 Fifth International Conference on Digital Information and Communication Technology and its Applications (DICTAP)*. IEEE.
- [11] Rodriguez, E. T. and Tamis-LeMonda, C. S. (2011). Trajectories of the home learning environment across the first 5 years: Associations with children's vocabulary and literacy skills at prekindergarten. *Child Development*, 82(4):1058–1075.
- [12] Shaik, F. A., Koshti, D., Gupta, A., Kalla, M., and Sharma, A. (2023). A comprehensive review and open challenges on visual question answering models. *AiBi Revista de Investigación, Administración E Ingeniería*, 11(3):126–142.
- [13] Su, W., Zhu, X., Cao, Y., Li, B., Lu, L., Wei, F., and Dai, J. (2020). Vi-bert: Pre-training of generic visual-linguistic representations. *ArXiv:1908.08530 [CS]*.
- [14] Summaira, J., Li, X., Shoib, A. M., and Abdul, J. (2022). A review on methods and applications in multimodal deep learning.
- [15] Uehara, K., Tejero-De-Pablos, A., Ushiku, Y., and Harada, T. (2018). Visual question generation for class acquisition of unknown objects. In *Proceedings of the European conference on computer vision (ECCV)*, pages 481–496.
- [16] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Brain, G., Research, G., and Polosukhin, I. (2017). Attention is all you need.
- [17] Wang, H. and Ge, W. (2024). Q&a prompts: Discovering rich visual clues through mining question-answer prompts for vqa requiring diverse world knowledge.
- [18] Xiaoqing, Z. and Junjun, H. (2021). Research on visual question answering based on deep stacked attention network. *Journal of Physics: Conference Series*, 1873(1):012047.